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Federated learning design and functional models: survey

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Abstract
Federated learning is a multiple device collaboration setup designed to solve machine learning problems under framework for aggregation and knowledge transfer in distributed local data. This distributed model ensures the privacy of data at each local node. Owing to its relevance, there has been extensive research activities and outcomes in federated learning with expanded applicability to different areas by the research community. As such, there is a vast research archive made available by the community with research work and articles related to the various aspects of federated learning such as applications, challenges, privacy, functionalities, and design. With respect to the function and design of federated learning, client selection, aggregation, knowledge transfer, management of distributed data (Non-IID), Incentive of data and communication cost are of paramount importance. Any effective design of federated learning requires these aspects to be well considered. There are numerous survey articles found among the available literature that focus on its application and challenges, opportunities, data privacy and protection, as well as on federated learning on internet of things, federated learning on edge computing, etc. In this paper, a review of the available literature on the various elements of design and functionalities in federated learning has been carried out with an aim to lay emphasis on the important challenges and research opportunities. More specifically, this work has endeavored to understand and summarize the various functional methods available, along with their techniques and goals. Additionally, it has strived to get a bird's eye view of how various functions and designs of federated learning have been used in applications, and how it has helped uncover challenges and promising research directions for the future.

Keywords Survey · Federated leaning · Design and function · Node selection · Aggregation · Non-IID management · Optimal communication cost

Extended author information available on the last page of the article

1 Introduction

Given the enormous and unprecedented amount of distributed data available today, it is no surprise that Federated Learning (FL) has received considerable attention across a range of applications. Consequently, FL has been utilized in applications across various sectors such as transportation, health, insurance and banking, Industrial and IoT-based applications, etc. FL is associated with a high degree of reliability while managing massive amounts of data and applications and this has generated a lot of research interest in pertinent areas. Among them, the primary areas of research surrounding FL have been focused on the technical challenges, the machine learning models, system modelling and design, privacy and security, data management, resource management, to name a few. In addition to such work, a number of related surveys addressing various aspects of FL have also been conducted and referred to since the very beginning. A summary of previously published studies, along with the categories of related surveys, and their focus points have been presented in Table 1.

As can be observed, the above surveys have addressed various aspects of FL. To elaborate, the surveys in Li et al. (2020a, b), Mammen (2021), Abdulrahman et al. (2021a), Zhang et al. (2021a), Rahman et al. (2021) were focused on the various challenges, opportunities, and future directions of Federated Learning. The surveys in Xia et al. (2021), Lim et al. (2020), Nguyen et al. (2021c), Imteaj et al. (2022), Aledhari et al. (2020) discussed edge computing, the Internet of Things (IoT), enabling technologies, solutions, and opportunities. The surveys in Li et al. (2021a), Mothukuri et al. (2021), Lyu et al. (2020), Kulkarni et al. (2020), Blanco-Justicia et al. (2021), Ghimire and Rawat (2022), Yin et al. (2021), Ouadrhiri and Abdelhadi (2022), Li et al. (2022b) discussed the areas of data protection, privacy, security, cybersecurity, secure distributed Learning, and various FL techniques. In a similar vein, Kholod et al. (2020) presented a survey of open-source FL frameworks for IoT, and the surveys in Liu et al. (2022b), Zhu et al. (2021a) focused on the structure of FL using distributed machine learning and neural architecture. The surveys in Yu et al. (2022), Zhu et al. (2021b), Jin et al. (2020), Prayitno Shyu et al. (2021) addressed the management of labeled data, non-IID, healthcare data, and data mining. Similarly, the surveys in Prayitno

Table 1 Summary of existing surveys

S. no.	Reference no.	Categories of surveys and focus points
1	Li et al. (2020a, b), Mammen (2021), Abdulrahman et al. (2021a), Zhang et al. (2021a), Rahman et al. (2021)	Implementation Challenges and Future prospects of FL
2	Xia et al. (2021), Lim et al. (2020), Nguyen et al. (2021b), Imteaj et al. (2022), Aledhari et al. (2020)	FL for Edge Computing and IoT
3	Li et al. (2021a, 2022), Mothukuri et al. (2021), Lyu et al. (2020), Kulkarni et al. (2020), Blanco-Justicia et al. (2021), Ghimire and Rawat (2022), Yin et al. (2021), Ouadrhiri and Abdelhadi (2022)	Security, privacy, and Blockchain of FL
4	Kholod et al. (2020)	Open-Source FL frameworks for IoT
5	Liu et al. (2022b), Zhu et al. (2021a)	Distributed machine learning and Neural architecture
6	Yu et al. (2022), Zhu et al. (2021b), Jin et al. (2020), Prayitno Shyu et al. (2021)	Fl for labeled data, Non-IID and Data Mining
7	Prayitno Shyu et al. (2021), Zheng et al. (2022), Zhou et al. (2021), Nguyen et al. (2022b), Antunes (2022), Pfitzner et al. (2021), Liu et al. (2021b)	Applications of FL

Shyu et al. (2021), Zheng et al. (2022), Zhou et al. (2021), Nguyen et al. (2022b), Antunes (2022), Pfitzner et al. (2021), Liu et al. (2021b) focused on the various applications of FL, especially in consideration to smart cities, healthcare, industrial IoT, etc. The summary above demonstrates the varied perspectives from which the surveys with respect to FL have been conducted over time.

The primary motivation of the survey presented in this work is the system model and the design of a federated Learning system. The predominant system models and design functionalities of FL have been described in this survey. In general, the most significant functionalities of the system model and design include client selection, aggregation and optimization, communication cost, knowledge transfer, data management, and incentives. These functionalities have been associated with different issues with respect to the system's design and the development of various applications. This survey undertakes an in-depth inspection of various challenges with regards to node selection, aggregation schemes, and knowledge transfer. In this context, Sawsan Abdulrahman et al. (2021a) have described the classification and functionalities of the model and design of FL, as has been shown in Fig. 1.

In this survey, three main functionalities in terms of contributions, issues, and future directions have been addressed, more specifically: client selection, node dropping, aggregation and optimization, knowledge transfer from client to global node and global node to client node, and data management (IID and Non-IID). The client selection is a process of selecting relevant nodes to start the training using local data. The selected node then acquires the current model and waits for the next model. Node dropping is the dropping of too-slow learning nodes from the cooperative machine learning models.

The aggregation and optimization model is a global node that provides an effective, high-quality model for the learning rounds. The knowledge transfer is a pre-trained model reused among the collaborative clients, with which the learning information of the clients is shared with the global node. The different characteristics and general overview of the client selec-

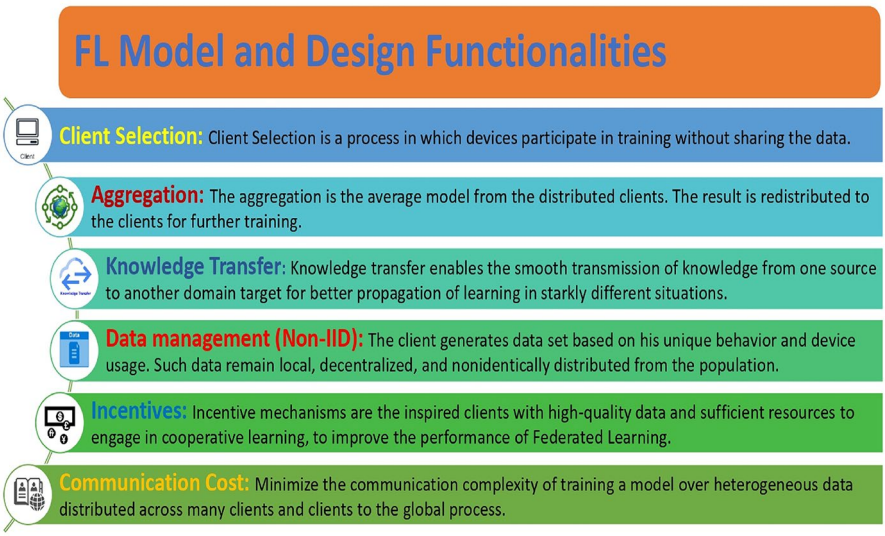


Fig. 1 Functionalities and design requirements of federated learning

tion process (Nishio and Yonetani 2019) from registration to client selections has been demonstrated in Fig. 2. The various steps involved in federated learning are as follows:

i. Initialization of the participant devices.
ii. Request sent to the resources for participation.
iii. Client Selection
iv. Distribution of the parameters.
v. Updating and uploading of the new parameters, and finally.
vi. Aggregating the transferred weights and parameters.With respect to the high requirements for the designing, collaboration, and applications of the FL system model and design, the issues that have emerged as the most pertinent for research with an eye to emerging technologies are client selection, aggregation and optimization, Knowledge transfer, and data management. In this context, the benefits and significant contributions of this paper compared to previously conducted surveys in the field can be summarized as follows:

1. The previous surveys (Li et al. 2020a, b, 2021a, 2022b; Mammen 2021; Abdulrahman et al. 2021a; Zhang et al. 2021a; Rahman et al. 2021; Xia et al. 2021; Lim et al. 2020; Nguyen et al. 2021c; Imteaj et al. 2022; Aledhari et al. 2020; Mothukuri et al. 2021; Lyu et al. 2020; Kulkarni et al. 2020; Blanco-Justicia et al. 2021; Ghimire and Rawat 2022; Yin et al. 2021; Ouadrhiri and Abdelhadi 2022; Kholod et al. 2020; Liu et al. 2021b, 2022b; Zhu et al. 2021a, b; Yu et al. 2022; Jin et al. 2020; Prayitno Shyu et al. 2021; Zheng et al. 2022; Zhou et al. 2021; Nguyen et al. 2022b; Antunes 2022; Pfitzner

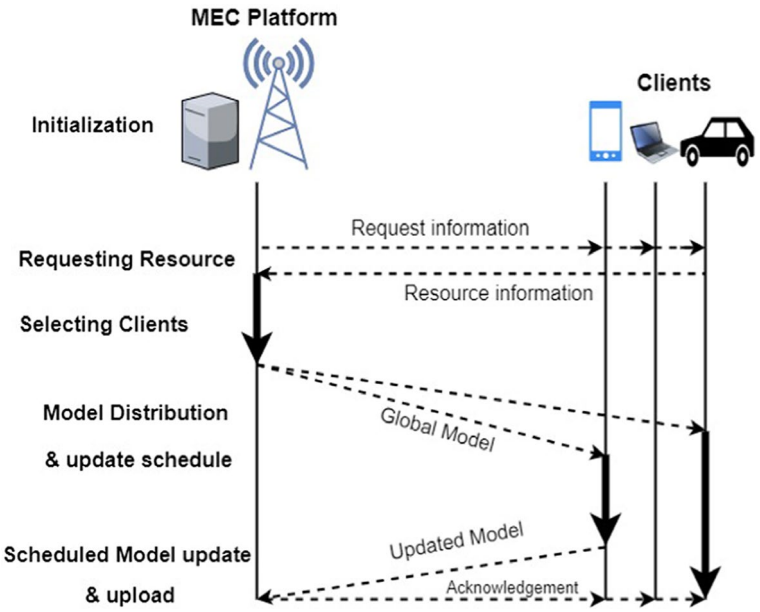


Fig. 2 Client selection process

et al. 2021) have mostly described FL challenges and opportunities, open-source frameworks, privacy, as well as different applications and emergent technologies such as edge computing and IoT. In contrast, this survey focuses specifically on the FL system model and design and the corresponding issues related to the various functionalities of the FL model and design.

2. Further elaboration is made on functionalities such as different client/node selection, and dropping node approaches with respect to various objectives, parameters, and characteristics are briefly described and compared.
3. In-depth analysis is provided of different aggregation and optimization methods in terms of various parameters and knowledge transfer from aggregation methods to client briefly discussed.
4. The different approaches to knowledge transfer methods, client data management, and cost management methods are analyzed and briefly described.
5. Finally how federated learning functionalities are used in different applications and the future direction of the research in terms of designing effective federated learning are presented.The rest of the work has been presented as follows. Section 2 compares and examines node selection and node dropping methods. Section 3 provides a comparative analysis of the various aggregation and optimization methods. Section 4 provides elaboration on knowledge transfer methods, and Sect. 5 describes data management on the client side. Section 6 presented Incentive management methods, and Sect. 7 described communication Cost reduce Methods. Section 8 presents how federated learning is applied to various applications. Finally, Sect. 9 provides a summary of issues and directions for future research related to the design of effective methods in federated learning, and Sect. 10 provides the conclusion.

2 Node selection and dropping methods

Node Selection Methods Client selection is a typical process in Federated learning where initially the device in question is charged and idle. All the registered edge devices are connected and associated to the server or global node to receive the global model and to transfer the communication accuracy and collected features and weights. The different characteristics and general overview of the client selection process (Nishio and Yonetani 2019) from registration to client selections has been demonstrated in Fig. 3.

In this section that elaborates upon the client selection process, different client selection methods and related feedback have been summarized. The different client selection methods have been presented in the references (Nishio and Yonetani 2019; Yoshida et al. 2020; Li et al. 2022g; Tang et al. 2022; Che et al. 2021; Xu et al. 2022b; Zhao and Joshi 2022; Cho et al. 2020; Larsson et al. 2021; Hosseinzadeh et al. 2022; Wang et al. 2021b; Luo et al. 2021b; Zheng et al. 2021; Tang et al. 2021; Cha et al. 2022; Wu and Wang 2022; Xu et al. 2021b; Lai et al. 2021; Huang et al. 2022; Abdulrahman et al. 2021b; Rai et al. 2022; Malandrino and Chiasserini 2021; Shin et al. 2020) while the different goals, datasets for the implementation and the metrics for the evaluation have been summarized in Table 2. In this case, the FedCS (Nishio and Yonetani 2019) was chosen as the primary client selection method and the main goal of this method is to actively select the client based on the resource condition.

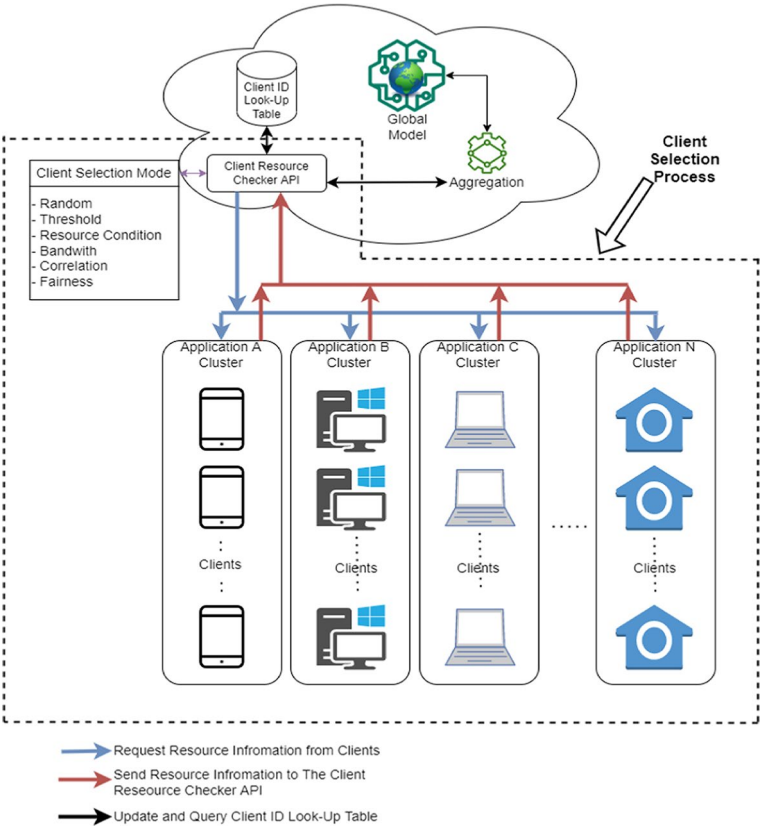


Fig. 3 Client selection process

The selection was performed using random criteria and the main drawback of this work was not considering dynamic scenarios and fluctuating dynamic updating. To address this, the multi-armed bandit (Yoshida et al. 2020) method was used to reduce the exploration and the exploitation and trade off in the mobile network. This method helps manage uncertainty caused by a huge amount of data, and one of the main advantages of this method is the reduction in learning time and is recommended in future scenarios for reducing resource usage in fluctuating FL operations.

The Power-of-Choice (Cho et al. 2020) model provides the guarantee of federated learning while managing communication, computational limitations as well as the heterogeneous data. In this method, the client selection is performed based on the random selection criteria. Additionally, the Power-of-Choice (Xu et al. 2022b) method has been used for managing biased client selection and has received better convergence rates compared to the random client selection method. Better performance in terms of faster convergence and more consistent management is made possible using methods like arbitrary client selection (Wang et al. 2021b) and adaptive client selection (Luo et al. 2021b) by assessing and updating probabilities pertaining to the client relationships. Using these types of client selection can help manage unknown parameters used for solving complex problems. Another approach based on client selection trade-off, where energy consumption is reduced with the help of a

Table 2 Summary of client selection methods

S. no.	Client selection models	Goal	Enviornment and dataset	Evaluation parameters
1	Fed CS-Client Selection Nishio and Yonetani (2019)	Client Selection based on client resource conditions	MEC, ML Tasks, MNIST, CIFAR-I0	Accuracy CIFAR-I0 (0.54%), MNIST (74%)
2	MAB-based Client Selection Yoshida et al. (2020)	Client Selection based on the rich throughput and computation cost	MEC, ML Tasks, CIFAR-I0	Learning Rate and Communication round accuracy
3	Power-of-Choice based Client Selection Cho et al. (2020)	Guarantee of FL training with a biased client selection method	DNN, FMNIST	Communication Round Accuracy (71.2% to 76.5%), Convergence speed, Global Loss
4	Arbitrary client sampling probabilities Luo et al. (2021b)	Reduced convergence time	MNIST and EMNIST	Convergence time, Communication cost and Accuracy
5	FedAECS Zheng et al. (2021)	Energy consumption management and balancing the trade-off between accuracy and cost in Edge client learning	MATLAB- Simulation	Energy Consumption and Accuracy
6	FedGP Tang et al. (2021)	Correlation-based client selection strategy	FMNIST and CIFAR-10	Communication Overhead
7	FedCor Tang et al. (2022)	Correlation-based client selection strategy	FMNIST and CIFAR-10	Communication Cost
8	Multi-Arm Bandit (MAB) Larsson et al. (2021)	Dynamic selection of clients for improved overall accuracy based on the base learning	MNIST	Communication Cost
9	PyramidFL Li et al. (2022g)	Achieving a higher final model performance (i.e., time-to-accuracy)	Multiple dataset is used	Accuracy, Clock Time
10	FedCorr Xu et al. (2022b)	Identification of noisy clients and separate measurement of all clients to find incorrect labels based on sample losses	CIFAR-10/100	Communication Test Accuracy
11	DRFL Zhao and Joshi (2022)	Biased Client Selection to encourage fairness and adjust the wait dynamically	FMNIST	Fairness
12	Distributed Client Selection Hosseinzadeh et al. (2022)	Client Devices for minimizing overall cost	CIFAR-10, FMNIST, and MNIST	Communication cost, Trade-Off
13	Fuzzy logic -Client selection Cha et al. (2022)	Client selection based on the quantity of samples, throughput, computational capabilities, and sample freshness	CIFAR-10	Communication Round Accuracy
14	FedPNS Wu and Wang (2022)	Selection of nodes that propel faster model convergence	CIFAR-10, CIFAR-100	Accuracy over Communication Round
15	Oort Lai et al. (2021)	Use of guided participant for selection and performance based on the time to accuracy and for quick training of clients	Google Speech, OpenImage and etc	Communication Rounds and Accuracy
16	E3CS Huang et al. (2022)	Boosting of convergence speed	EMNIST-Letter and CIFAR-10	Communication Rounds and Accuracy
17	FedMCCS Abdulrahman et al. (2021b)	Multi-Criteria Approach for client selection based on memory, timing, energy, and client resources	NSL-KDD	Communication Rounds and Accuracy

heuristic-based method is called FedAECS (Zheng et al. 2021). Similarly, the correlation-based Gaussian process (FedGP) (Tang et al. 2021) and the correlation-based client selection process (Fedcor) (Tang et al. 2022) used for the management of heterogeneous help to reduce communication overheads and communication costs. The CMFL Che et al. (2021) framework, on the other hand, used for random selection, in which two different selection strategies are used for managing the robustness and convergence of the selection strategies, results in faster convergence and better selection performance. The automated collaborator selection strategy for the management of non-iid data performed using multi armed bandit (MAB) (Yoshida et al. 2020) method has also been used to manage the overall benefits of federated selection.

Another improved federated training approach based on fine grained selection (Li et al. 2022g) is called the pyramidFL framework and has been used to improve the time-to-accuracy performance of federated learning. With the help of this framework, the local data efficiency is seen to improve, and the communication cost is reduced, while overall improving end-to-end performance. Another framework, the Fedcorr (Xu et al. 2022b) multi-stage selection framework, used for managing noise of local clients while independently managing all the clients, has been noted for increasing the fairness of the federated learning process by adjusting the client selection strategies and overall improving the efficiency of the results. The DRFL (Zhao and Joshi 2022) method is used to improve the weight accorded to each client and thereby increase the fairness of the result. In distributed client selections, it is an inherent challenge to address the problem arising out of the the trade-off between communication and cost.

The Distributed Client Selection (DCS) (Hosseinzadeh et al. 2022) method is used to alleviate this problem and using this, about 32.67% of the overall cost is seen to be reduced. In another approach, a fuzzy logic (Cha et al. 2022) based client selection method has been used in vehicular networks for managing heterogeneous data and network throughput. In this fuzzy logic-based client selection, multiple factors including network throughput is used for client selection. For achieving faster convergence of non-iid data, FedPNS (Wu and Wang 2022) node selection has been used, where the selection probability of each node is changed dynamically depending on the optimal aggregation. In another scenario, the MAB method (Xu et al. 2021b) has been used with a greedy algorithm to achieve trade-offs that may enhance suitability for optimization. This method has been observed to minimize communication delays, thus reducing training latency. In order to circumvent the poor system efficiency usually associated with random client selection, researchers have introduced a guided participant selection method called Oort (Lai et al. 2021).

Another method for avoiding random selection is the multi-criteria approach that is used for selecting the client in FL. Using this method, the number of communication rounds is reduced, while maximizing the number of clients and reducing the number of discarded client sessions at the same time. Use of this model has led to improvements in model accuracy and has helped train clients quickly. Volatile client selection methods have led to reduced performance in a Federated Learning system, affecting the training efficiency as well as fairness. To avoid this problem, a stochastic client selection (E3CS) scheme (Huang et al. 2022) has been used, which has resulted in a fixed accuracy as well as faster convergence. In scenarios with imperfect data, an irrelevance sampling method has been proposed for client selection (Rai et al. 2022), which generates an irrelevance score that considers the client characteristics. This method has resulted in faster convergence as well.

In the distributed large-scale neural network used, multiple edge nodes were used for local updates. To manage this scenario, a secure node selection method based on a blockchain-based random client selection method was introduced. Using this method resulted in improved privacy (Qammar et al. 2023; Li et al. 2020a). similarly, in a distributed IoT device selection, a resource-constrained based node selection method called the DDQN-trust solution uses random selection techniques. This DDQN method was integrated with various federated averaging methods such as FedAvg, FedProx and FedSGD (Rjoub et al. 2024). Another technique, deep reinforcement learning method-based node selection, was introduced for heterogeneous data in a distributed client selection. This client selection was performed based on the client quality record for a dynamic environment using the Markov decision process and CS-A2C (Li et al. 2024). For edge computing in the IoT environment, the deep reinforcement learning-based node selection was introduced, and heterogeneous data was utilized in the edge devices. Using this method, better privacy was achieved (Yan et al. 2023). The grouped similar techniques based on the selection strategies such as random, resource constraints, bandwidth, etc presented in Table 3.

Dropping of Nodes Node dropping refers to the temporary or permanent removal of certain clients from training rounds. This can occur for multiple reasons, such as limited resources, network instability, and strategies adopted to improve efficiency and scalability. The main task of Federated Learning is cooperatively training the machine learning model using a central server to coordinate with multiple participants or clients. These servers may occasionally drop slow nodes from the learning process. The authors of Malandrino and Chiasserini (2021) suggested some key issues that may lead to the dropping of nodes from the learning process as follows:

i. Insufficient quality of data.
ii. Slow learning.
iii. Low bandwidth.

Table 3 Summary of similar techniques based on the selection strategies

S. no.	Client selection strategies	Techniques	Impact analysis
1	Random client Selection	Power of -Choice (Xu et al. 2022b; Larsson et al. 2021), CMFL-framework, (Che et al. 2021), Oort (Lai et al. 2021)	The main impact of random client selection is inefficiency while using unreliable clients, potential bias, and communication overheads due to increased communication rounds
2	Bandwidth Client Selection Method	Fuzzy logic based client Selection (Cha et al. 2022), MAB method (Xu et al. 2021b), Stochastic Client Selection (Huang et al. 2022), FedPNS (Wu and Wang 2022)	The impact of bandwidth client selection is limited client pool, network variability, implementation complexity, and inefficiency in resource utilization
3	Resource Condition based Client Selection	FedCS (Nishio and Yonetani 2019), Multi-armed bandit (Yoshida et al. 2020), PyramidFL (Li et al. 2022g), FedMCCS (Abdulrahman et al. 2021b)	The impact of resource condition client selection is reduced fairness, biased data representation, over-reliance on high resource clients, and inefficiency in diverse networks
4	Correlation-Based client selection	FedGP (Tang et al. 2021), FedCor (Tang et al. 2022), FedCorr (Xu et al. 2022b)	The impact of correlation-based client selection is reduced fairness, overfitting, data representation issues, computational complexity, etc

iv. lack of updates and a high degree of noise.Some other reasons that may be considered before dropping nodes are Non-homogeneous client data and Node with different capabilities. The authors of Wang et al. (2023) and Li et al. (2022c) used node drop strategies for data privacy in a decentralized federated learning system. Wang et al. (2023), used zk-SNARK techniques to enhance the trustworthiness of the node, using node dropping and joining strategies. Similarly, Li et al. (2022c), proposed a framework called GALAXY to achieve privacy in a cooperative multiple-user framework, permitting a certain number of fog nodes to cooperate and keeping the system robust with a certain number of fog nodes dropping out simultaneously. Using this GALAXY framework led to better scalability, higher efficiency, and reduced resource overheads in fog environments. For the large classes of clients, FL-FDMS was introduced in the distributed environment for data similarity. In this FL-FDMS, the clients are dropped out of the distributed environments (Wang and Xu 2022) using the similarity of data.

Implications and Robustness of node dropping There are various positive and negative implications of the learning process in federated learning. Reducing communication costs and overheads, proper resource utilization, better model convergence, coordination of clients, fault tolerance, managing attacks from the suspected nodes, and adaptive learning from the relevant and effective resources are some of the positive implications and areas of robustness. Data representation loss from unique and important clients, potential bias due to certain types of clients, reduced diversity due to a less diverse training dataset, and dependency on only high-performance clients to significantly reduce the training process of FL are some of the negative implications of node dropping.

Discussion Most of the client selection methods employed till date have been based on random selection, resource constraint selection, and bandwidth-based client selection patterns as a means of selecting clients. These types of client selection patterns typically give rise to issues such as increased communication costs, inadequate quality of data and lack of effective clients since in such selection methods, frequent updating of data is not given due consideration.

3 Aggregation methods

The main goal of FL is generating an effective global model with the help of models learned on the distributed local data of the participating clients. The high quality of data of the learned client models are then aggregated or combined to generate a new kind of model called the aggregation model. In general, both aggregation algorithms and rank aggregation algorithms are applied with different application such as internet applications, meta-data analysis, etc. The algorithms generally used for the aggregation (Lin 2010) are mainly heuristic and stochastic in nature. There are usually two layers of aggregation incorporated in federated learning, such as local device aggregation and federated or cross device aggregation (McMahan et al. 2017).

Local Aggregation Local aggregation refers to the multiple samples owned by individual clients. It uses the model parameters while sequentially using the local parameters as well. It considers computer accuracy, average loss, and other metrics for local iterations

with the help of local streams Federated Learning | TensorFlow Federated. https://www.tensorflow.org/federated/federated_learning.

Federated Aggregation Federated aggregation is also referred to as cross-device aggregation. This refers to aggregation across multiple connected devices and uses model parameters, averaged client aggregations as well as other metrics from the model in order to arrive at a global model Federated Learning | TensorFlow Federated. https://www.tensorflow.org/federated/federated_learning. The diagrammatic representation of a Federated Aggregation structure is illustrated in Fig. 4. Researchers in the area have considered different federated aggregation methods over time to arrive at effective models.

Initially, federated averaging Yao et al. (2020) was introduced using a stochastic gradient optimization for averaging five different models using four datasets. This was the base model for federated aggregation. Konečný et al. (2016) subsequently introduced unevenly distributed large nodes for training the centralized model using high quality data. This process was called "federated optimization" or "algorithm." In this method, features, number of nodes, and communication rounds were used for evaluation. Continuing the thread, Wang (2017) conducted further research and proposed an asynchronous protocol called CO-OP for training the local model with the help of newly generated samples and merging the local samples into global models. In this work, model accuracy and communication overhead parameters were considered for evaluation.

These references (Konečný et al. 2016; Wang 2017; Guha et al. 2019) point to the base aggregation models, based on which future models were developed. Using the one-shot federated learning (Guha et al. 2019) method, the entire participant devices' knowledge was shared within a single round of communication. But the average relative gain was found to be only 51.5%. The periodic average (Haddadpour and Mahdavi 2019) was calculated using different data distributions, the unbiased gradient and sample schema, and the polyak-lojasiewkz condition, which provided the best convergence rates. To manage corrupted updates, a robust aggregation method (Pillutla et al. 2022) was used with the help of a geometric median (GM). The author, Xie et al. (2021), presented a discrepancy in the data distribution, and proposed a multi-center federated aggregation using different global nodes.

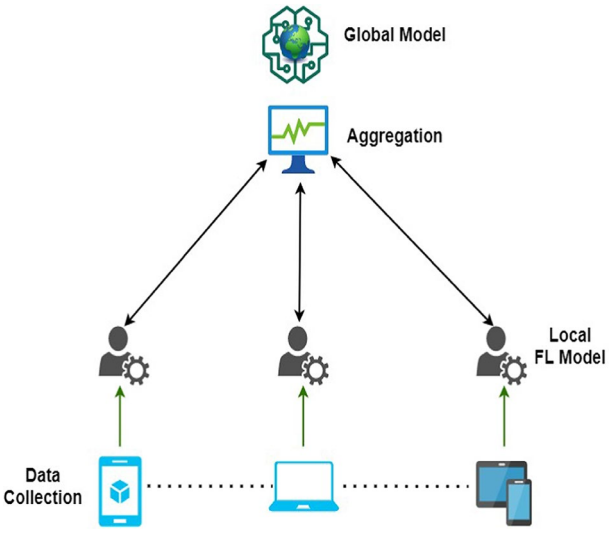


Fig. 4 Federated aggregation structure

To avoid end-to-end training latency, the author of Jiang and Hu (2020) proposed the Fed-PGA mechanism to exchange the particle gradient mechanism and to speed up the overall communication process and to reduce communication overheads. Time and accuracy were the two evaluation parameters that were considered in the evaluation. The IDA (Yeganeh et al. 2020) aggregation method was used to robustly aggregate clients using the parameters and the FEDBE (Chen and Chao 2021) model was used to robustly aggregate the performance using a Bayesian ensemble model. This model was used to send single round trains to the clients and multiple rounds were avoided.

To avoid the scenario of different clients having different local models, a model called MHAT (Hu et al. 2021) was proposed to make an auxiliary model to extract and update the real information for aggregation and reduce the communication rounds. The author of Wang et al. (2020) introduced a matched average aggregation method using hidden elements such as the hidden layers of the CNN and LSTM algorithms and connected layers-wise matching was then performed. Using this method, the communication overhead and rounds were reduced. To avoid unnecessary server updates and scalability issues, the authors of Nguyen et al. (2022d) proposed a model called FedBuff. In this model, before updates to the server were made, 'K' client updates were stored in a sequential buffer, before being updated to the server. To increase fairness and robustness, the authors proposed a model called Ditto (Li et al. 2021c).

In this framework, a single framework for multi-tasking learning was performed. In this work, two tasks, such as global and local model objectives, were used to incorporate and regulate the optimal global model. The authors of Elkordy and Avestimehr (2021) provide a method for sequential aggregation using heterogeneous quantization. Using this method, edge users adjusted the quantization to the resources available and succeeded in providing better accuracy and lower communication costs. In another approach, to tackle the issues of corrupted data, a robust aggregation method (Pillutla et al. 2022) was used, in which the aggregation was performed in two ways, viz. a one-step aggregation in the global model and another on device personalization aggregation. Due to the multiple local updates and consistency between the targeted distribution, the authors' proposed unbiased gradient variant (Yao et al. 2020) aggregation method was used, and using this method, faster convergence and higher accuracy were achieved. The author of Palihawadana et al. (2022) proposed a FedSim aggregation model for local and global aggregation. In this method, at first, similar gradients were clustered using local aggregation and then globally aggregated for better convergence and variance. In a similar vein, the quality of the gradients was provided by the analog gradient aggregation (Guo et al. 2021) for learning over wireless using SGD. A summary of the above aggregation methods has been presented in Table 4.

For securing the aggregation process, HeteroSAg was introduced (Zhang et al. 2023b) using heterogeneous quantization. The edge users adjusted their quantization to available resources using HeteroSAg and achieved the best trade-off between accuracy and communication costs. Similarly, for secure aggregation and robustness against client dropping out, it was introduced in the federated environments (Eltaras et al. 2023). To address the security challenges in a centralized aggregation, a trustworthy aggregation model called DeTA (Cheng et al. 2024) was introduced, using the DETA multi-aggregation method used in the FL ecosystem. The model was aggregated using this approach without any alteration. This resulted in improved privacy. Similarly, for better privacy in the distributed environment, the DOTFL method was introduced using a neural network. Malicious activities were

detected in the FL system using the DOTFL. To personalize FL, a local federated learning-based method called FedALA was introduced to impair the generalization of the global model. Another important issue in FL is limited bandwidth and energy in the distributed environment (Zhang et al. 2023a). To address this, particle aggregation was introduced, and the model is called PMA-FL in the wireless environment (Chen et al. 2024). Using the PMA-FL, the features extraction part was shared for prediction and the learning process.

Relationship Between Node Selection and Aggregation The efficiency of FL heavily depends on how well the node selection methods and aggregation methods have been integrated. These two functions need to be carefully synchronized to manage communication costs and data integrity in the system. The main node selection methods that have been considered are random selection, importance-based selection, resource-awareness based selection, specific criteria-based selection and reward-based selection. The random selection method is fair but not efficient, and aggregation without efficiency would lead to increased communication costs and efficiency. The importance based, resource-awareness-based and reward-based selection methods demonstrate more effective selection with regards to clients in a distributed environment. This produces better aggregation results globally with reduced communication costs in terms of measuring the quality of data, measuring the capability of the training data, and encouraging better participation. The specific criteria-based selection creates data groups based on certain criteria and the selection is done proportionally from each group of data. This helps in dealing with data heterogeneity in disparate environments.

Furthermore, by considering a local aggregation methods, data quality is improved since it results in locally updated and similarly clustered data that is globally aggregated, resulting in reduced communication costs. Therefore, resource-awareness based selection, specific criteria-based selection and reward-based selection methods, combined with local aggregation methods provide reduced frequency of communication, and result in a better performance model in collaborative environments (Alferaidi et al. 2022). Careful integration of node selection and aggregation methods in federated learning is crucial for optimizing communication costs and handling data heterogeneity. Thoughtful designing and aligning of these methods can help in increasing the efficiency and performance of federated learning systems (Wu and Wang 2022).

4 Knowledge transfer methods

Knowledge transfer enables the smooth transmission of knowledge from one domain (source) to another domain (target) for better propagation of learning even under starkly different situations (Li et al. 2020b; Yang et al. 2019a). The authors of Yang et al. (2019a), Saha and Ahmad (2021) categorized federated learning into three groups: horizontal, vertical, and transfer federated learning. In the case of horizontal FL, an overlap of certain features can be observed across the different nodes. Moreover, the same feature dimension is used across all nodes. In contrast, the features of a vertical FL include a vertical dimension. For example, a problem may have the same entities but different characters. In the case of federated transfer learning, the knowledge is transferred into a different domain altogether. Hence the federated knowledge transfer differs radically from the vertical and horizontal knowledge transfer.

Table 4 Summary of aggregation methods

S. no.	Aggregation models	Goal	Enviornment and dataset	Metrics
1	Federated Averaging McMahan et al. (2017)	Weighted Average of stochastic gradient descent	CIFAR-10	Communication Rounds and Accuracy
2	Stochastic Average Gradient (SAG) Yao et al. (2020)	Combining the different forms of SGD (Stochastic reduced gradient and Stochastic dual coordinates) with fast convergence of GD	Public Google+ post	Communication Round
3	One-shot federated learning Guha et al. (2019)	Updating the global model using a single round of communication	EMNIST, Sentiment140	–
4	FedProx Li et al. (2020d)	Modifying and updating the re-parameterization in the Framework	EMNIST, Sentiment140	Convergence
5	Robust Aggregation Pillutla et al. (2022)	Aggregation performed using FedAvg with a geometric median	EMNIST, Sentiment140	Accuracy
6	Multi-center aggregation mechanism Xie et al. (2021)	Consideration of data discrepancy while using multiple global models	FEMNIST	Convergence
7	IDA (Inverse Distance Aggregation) Yeganeh et al. (2020)	Reject the out of distribution of the model and then aggregate	CIFAR-10, F-MNIST	Communication Round Accuracy
8	FEDBE (Bayesian model Ensemble) Chen and Chao (2021)	Bayesian Ensemble Model to avoided Multiple communication rounds	CIFAR-10/100	Communication Round Accuracy
9	MHAT: Model-heterogenous aggregation training Hu et al. (2021)	Auxiliary model used to extract and aggregate real Information	MNIST	Convergence rate
10	FedMA: Matched Averaging Wang et al. (2020)	Layer-wise matching between hidden layers of client and global models	CIFAR-10	Communication Overhead and Accuracy
11	Buffered Asynchronous Aggregation Nguyen et al. (2022d)	The k client securely updated in the buffer and update in the server prior to any communication to the global model	Sentiment140, CIFAR-10	Communication Round and Convergence Analysis
12	Ditto Li et al. (2021c)	Achieving fairness by using the local objective and the global objective as the global mode	FEMNIST	Communication Round
13	RFA Pillutla et al. (2022)	Update of corrupted data using robust Aggregation methods	EMNIST, Sentiment140	Communication Cost
14	Unbiased Gradient Aggregation Yao et al. (2020)	Meta-data aggregation using Gradient Aggregation strategy	CIFAR-10	Convergence Rate and Accuracy
15	Analog Gradient Aggregation Guo et al. (2021)	Wireless aggregation for learning over air using SGD	Handwritten-digit	–

Table 4 (continued)

S. no.	Aggregation models	Goal	Enviornment and dataset	Metrics
16	FedSim Palihawadana et al. (2022)	Clustering of the local client aggregations and subsequent global aggregation	Recognition	Communication Round

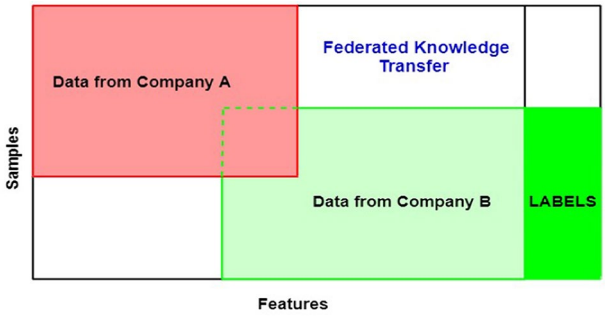


Fig. 5 Federated knowledge transfer (Yang et al. 2019a; Saha and Ahmad 2021)

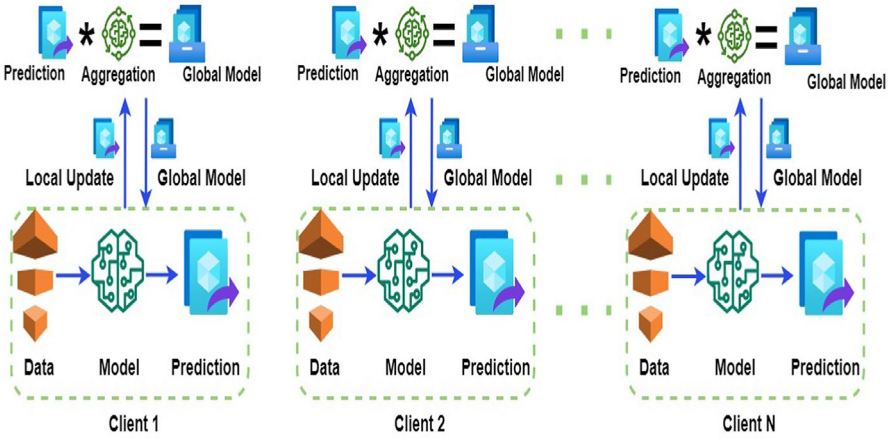


Fig. 6 Federated knowledge transfer steps

In this section, various works related to federated knowledge transfer have been presented. Federated knowledge transfer is widely used in applications across different verticals and areas such as industrial, healthcare, security, and user privacy, etc. Therefore, a fairly diverse set of applications are found to use knowledge transfer for sharing of features for better results. The diagrammatic representation of knowledge transfer in FL is presented in the Fig. 5. The process of step by step knowledge transfer representation shown in Fig. 6.

Further, different knowledge transfer techniques related to federated learning is presented along with their corresponding limitations. A summary of the different techniques, goals, implementation details and metrics has been presented in Table 5. The author of Li et al. (2022d) demonstrated a decentralized FL utilizing mutual knowledge sharing by using Def-KT, wherein the local client transferred their learning knowledge to other nodes. The combo and full average were used for averaging in the IoT system. In this work, the communication cost was seen to be high. In another instance, a group knowledge transfer was performed using the federated learning method called FedGKT (He et al. 2020). Using FedGKT, the resource constraint problem was addressed in the edge devices with knowledge transfer formulated using edge devices. Personalized federated learning was performed using parametrized knowledge transfer (KT-pFL). In KT-pFL, the personalized soft prediction was maintained by using local training and collaborating with different clients

Table 5 Summary of knowledge transfer methods

S. no.	Knowledge transfer method	Goal	Enviornment and dataset	Metrics
1	Decentralized FL—knowledge Transfer—Def-KT Li et al. (2022d)	Local clients to transfer the learned knowledge	MNIST, Fashion-MNIST and CIFAR-10	Classification Accuracy
2	Group knowledge Transfer—FedGKT He et al. (2020)	To tackle resource constraints using FedGKT	CIFAR-10, CIFAR-100	Computational efficiency and communication efficiency
3	Parameterized Knowledge Transfer Zhang et al. (2021b)	To maintain the personalized soft prediction and collaboration among the clients	EMNIST, Fashion-MNIST	Communication round and efficiency, Accuracy
4	Zero-Shot Knowledge Transfer-FedZKT Zhang et al. (2022)	To enable knowledge transfer across the heterogenous on -devices using FedZKT	MNIST, KMNIST, FASHION and CIFAR-10	Learning Curve or rates, Average Accuracy
5	Clustered Knowledge Transfer- PERFED-CKT Cho et al. (2021)	To transfer knowledge to clients with similar data distribution using clustered co-distillation	CIFAR100	Accuracy, Communication Cost
6	Cross-Device Knowledge (CDKT-FL) Transfer Nguyen et al. (2022a)	To enable CDKT-FL transferred knowledge across global knowledge and on-device	Fashion-MNIST, MNIST, CIFAR-10	Accuracy
7	Fed-ET-Knowledge Transfer Cho et al. (2022)	To use framework for training large servers using small and heterogenous models on clients	CIFAR10, CIFAR100	Communication Efficiency
8	Heterogeneous Transfer learning -SPATL Yu et al. (2021)	To enable transfer learning and features selection in heterogenous clients using local updates	FEMNIST and CIFAR-10	Communication Round and accuracy
9	FedFTG Zhang et al. (2022a)	To enable knowledge distillation using Training	CIFAR10, CIFAR100	Accuracy
10	FedBD Le et al. (2021), FedKD Wu et al. (2022)	To enable the exchange of soft targets between client and server	CIFAR-10	Accuracy

using similar dimensions. The communication round, efficiency and accuracy were considered for evaluating KT-pFL.

To avoid round communication, the authors of Li et al. (2020c) proposed a model called round optimal FL via knowledge transfer where optimal agnostic accuracy was increased and privacy loss was reduced. The authors of Zhang et al. (2022) proposed a model called zero-shot knowledge (FedZKT) transfer. This method proposed the transfer of knowledge across the heterogeneous models in a data-free manner in distributed devices. In another instance, the authors of Cho et al. (2021) presented a clustered knowledge (PERFED-CKT) transfer for similar data distribution in personalized devices using the clustered data distribution, reducing the communication cost, and increasing the accuracy. The authors of Nguyen et al. (2022a) proposed a cross-device knowledge transfer for global knowledge and on-device. This model was observed to successfully speedup and personalize the performance of the client model. In this work, the FedAvg algorithm was used for training and for minimizing the communication load when exchanging the models. To train the large server using small and heterogenous clients, authors of Cho et al. (2022) proposed the Fed-ET. Three consecutive steps were used to manage the Fed-ET such as local training and translator, weighted diversity regulation, and server transfer. Using this method communication efficiency was increased.

The authors of Yu et al. (2021) proposed a SPATL model for knowledge transfer and feature selection in heterogenous client using local updates. The authors of Zhang et al. (2022a) proposed a model called FedFTG for knowledge distillation to fine-tune global models using local models. The local model was transferred to the global model in heterogeneous scenarios. The authors of Le et al. (2021) proposed a soft target for the reference dataset model called FedBD. The knowledge distribution minimized the gap between server and local devices after output was received from the clients. The client updated the local model by training the reference and private data using this model for knowledge distillation. The authors of Wu et al. (2022) also proposed knowledge distillation for effective communication in federated learning. The FedKD model with an adaptive mutual distillation was used for effective collaboration and communication. The authors of Chang et al. (2019), Pei et al. (2022), Pan et al. (2021), Qi et al. (2022), Shi and Radu (2021), Zhao et al. (2020), Li et al. (2022f) used knowledge transfer to manage private data, malware detection, anomaly detection and dependency, as well as inversion attack using collaborative machine learning.

The traditional horizontal federated learning (FL) is affected by non-IID distributions. To improve the personalized private learning model, the proposed pFedKT framework (Yi et al. 2024) for private and global knowledge transfer is introduced. In pFedKT, two types of knowledge transfer are achieved using hypernetworks for transferring historical data into a new private model and contrastive learning for transferring global model knowledge into a private model for improved accuracy. Similarly, while the same horizontal federated learning protects the local private data of each participant, it is not sufficient for the learning model. To overcome this issue, FedKT (Wang et al. 2024a) uses fine-tuning and knowledge extraction from the early layers of processing. To achieve sustainable development, a new knowledge level introduces four sources: tasks, features, models, and algorithms (Wang et al. 2024b). It also introduces new federated continual learning for the transmission of knowledge. To share secure knowledge in a distributed environment, the KD-IS (Shi et al. 2024) framework is introduced for monitoring poor performance models.

Discussion It is evident from the above summary that the features and data characteristics in question have been are changing from client to client based on various criteria. In essence, in federated learning, managing of clients, effective learning, and updating of knowledge transfer methods need to evolve further based on specific criteria. As far as knowledge transfer is concerned, communication cost, communication rounds, balancing the clients and maintaining privacy during knowledge transfer, as well as managing the energy expenditure are the essential factors to be considered in knowledge sharing.

5 Data management

In federated learning, data collection and behavior property are based on the non-IID. Non-IID means non-independent and identical distribution. In the federated learning data, the collection of the random variables with independent and identical distributed random variables has the same probability and is mutually independent. The different data properties and characteristics are presented in Ma et al. (2022) but do not describe the different methods for managing non-IID data. The non-IID in the federated learning method manages data sharing, selection, enhancement, data updation, algorithm-based learning such as meta, multi-tasking, and lifelong learning, and classification of data based on similarity, knowledge distillation, and personalization. Figure 7 presents non-IID data management related to health care. In Fig. 7, the different clients have identical, and the data distribution is also different.

Different methods are present to manage the non-IID data in federated learning. A summary of the different techniques, goals, datasets, and metrics has been presented in Table 6. The authors of Briggs et al. (2020) present a method to improve the non-IID data training in local updates using the FL + HL method. In this method, the data is clustered based on the similarity of local data. The pairwise collaboration of similar data between the clients is managed using the FedAMP (Huang et al. 2021) method. The authors of Li et al. (2022e) present a comprehensive partitioning of different non-IID data case strategies to manage heterogeneous data distribution in different parties, and the authors of Zhao et al. (2022) present non-IID data user behavior in the 6 G wireless environment using federated learning. The authors Chen et al. (2020b) present an ASO-fed method for managing heteroge-

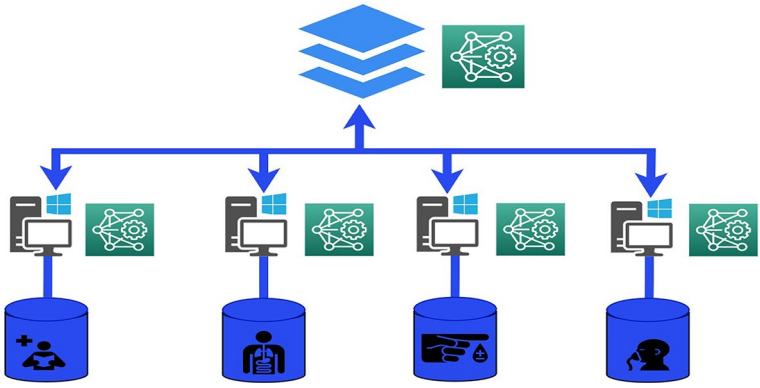


Fig. 7 Data management (non-IID)

Table 6 Summary of data management (Non-IID)

S. no.	Methods	Goal	Dataset	Metrics
1	FL+HC Briggs et al. (2020)	Similarity of Local data	MNIST	Communication Rounds and Accuracy
2	FedAMP Huang et al. (2021)	Pairwise collaboration of similar Data	MNIST, FMNIST	Accuracy of No of clients
3	ASO-Fed Chen et al. (2020b)	Managing Heterogeneity of devices and sampling Data	Fashion-MNIST	Accuracy, sample Rates, and F1 score
4	DS-FL Itahara et al. (2021)	Managed ambiguity of data sample, and lowering the training convergence	MNIST	Communication Rounds and Classifications accuracy
5	FedCD Kopparapu et al. (2020)	Dynamically group Clients with similar data	CIFAR-10	Communication Rounds and accuracy
6	PENS Onoszko et al. (2021)	Similar Data distribution and evaluate the training loss	CIFAR-10	Accuracy
7	FeDBN Li et al. (2021d)	Clint Data Normalization and features shifting before averaging	Altech10	Overall Compression and Accuracy
8	FedSA Chen et al. (2021a)	Ensemble learning used for Communication efficiency	Synthetic dataset	Accuracy
9	FedUFO Zhang et al. (2021c)	Manage inconsistency and feature divergence Issues of data.	MNIST and CIFAR-10	Accuracy.

neous devices and sampling data in edge devices. In this work volume of data, varying latency and availability issues are considered. The authors of Itahara et al. (2021) proposed the DS-FL method to exchange the output of local data, manage the ambiguity of data samples, and manage training convergence. Using this method reduces the communication cost and achieves better classification of similar data. FedCD (Kopparapu et al. 2020) dynamically groups similar data in different clients. Using the FedCD method, higher accuracy, fast convergence, and manageable overhead problems in federated learning. Similarly, the author of Onoszko et al. (2021) proposed a method called PENS and, using this work, similar data distribution in clients and evaluated the training losses in federated learning. The authors of Li et al. (2021d) proposed a method called FedBN for normalizing the client data and shifting the features of the data before aggregation. The authors of Chen et al. (2021a) and Zhang et al. (2021c) proposed a method called FedSA (Chen et al. 2021a) and FedUFO (Zhang et al. 2021c) to manage unstable, unreliable, inconsistentency, and feature divergence problems in distributed non-IID. Using these two methods, the learning performance is increased in federated learning. And also, the authors of Zhang et al. (2020) proposed a method to diagnose and detect arrhythmia using non-IID ECG. Using this method, medical institutions can collaboratively train and detect arrhythmia using federated learning. The optimized data management method introduced in the distributed environment is called ODMSM-FL (Ramani et al. 2024). This method achieves secure federated learning in the distributed environment using blockchain technology.

Discussion The data distribution with a non-IID and personalized FL has a number of potential issues. The major challenges with non-IID data distribution are divergence in local models for making aggregations into a global model, slower convergence rate due to non-IID data, biased global models due to certain data distribution and inconsistent contribution issues due to variability in the training process. Personalized federated learning provides

better privacy, with stronger privacy algorithms required. Beyond that, model personalization and globalization, resource constraints, scalability, and complexity of implementation are some of the potential issues in data distribution and personalized federated learning.

6 Incentives

Incentive mechanisms in FL refer to the improvement in the quality of data resulting in better accuracy with fewer iterations, thus leading to better overall efficiency and energy-saving measures. The authors of Abdulrahman et al. (2021a), Tu et al. (2022), Seon Hong et al. (2021) described such an incentive mechanism that would reward efforts towards achieving a higher quality of data. The authors of Seon Hong et al. (2021) provided an incentive mechanism related to the design of federated learning and chose to use game theory and action theory for designing the incentive mechanism. In order to manage false results on the local data and to determine the truthiness of the client, the authors of Liu and Wei (2020) decided to introduce a solution based on the Bayesian Nash equilibrium framework. Using this model, the real client truth was verified. The authors of Sun et al. (2022) proposed a static and dynamic design for building an incentive mechanism for federated learning using the real-time status of the network elements that assisted the design using FL. The authors of Richardson et al. (2020) proposed a game-theoretic scheme for finding the truthiness of data and for reducing data redundancy. Similarly, the authors of Chen et al. (2020a) used a gamified research tool to manage the high-quality data in the FL eco-system. The authors of Wang et al. (2021a) presented a fair and robust federated learning (FRFL) to address the incentive mismatch and free-rider attack for UAVs. The authors of Yu et al. (2020) proposed an FLI (Federated learning incentivizer) to maximize the collectiveness of the data and to minimize data inequalities in FL. In the context of competitive situations, the role of incentives in federated learning cannot be overstated. Clients are motivated to participate in information gathering, which in turn reduces the cost and resources required. This is achieved through three key components (Sim et al. 2024)in the shared environment: fair contribution of client data, client selection to maximize the global model, and reward-based allocation to clients. The higher rewards provide better outcomes. The summary of the incentive methods is presented in Table 7.

Discussion High-quality client data is required to manage fast convergence and effective global models (Abdulrahman et al. 2021a). A practical incentivizing approach is required for managing the client's data and to this end, the verification of the client's data density, data update, and the influence of new data must be considered in building new incentive approaches.

7 Communication cost

Continuous communication in federated learning requires an extensive number of communication rounds between the client and server due to the sizeable data and number of clients. This necessitates the use of effective algorithms and models to reduce the cost between the client and the global model (Kang and Ahn 2021). To that end, a number of approaches have been used to decrease communication rounds and improve performance. The Fig. 8 repre-

Table 7 Summary of incentive methods

S. no.	Incentives model	Goal	Dataset	Metrics
1	Hypothesis Elicitation approach Liu and Wei (2020)	Enable incentive verification using Bayesian Nash Equilibrium Approach	CIFAR-10, MNIST	Misreport Rates
2	Incentive mechanisms for FL Sun et al. (2022)	Utilize a Stackelberg game to design incentives for FL	MNIST	Accuracy and Energy Consumption
3	Budget-Bounded Incentive for FL Richardson et al. (2020)	Utilize game-theoretic scheme for finding truthfulness and redundancy of data	–	Influence of Data
4	FL with Fair Incentives Wang et al. (2021a)	Use a fair and robust FL scheme for crowdsensing	MNIST	Data size, average Transmission speed, and utility of task
5	Fairness-aware Incentive -FLIf Yu et al. (2020)	Minimize Data inequality	–	Data Quality and Total Revenue

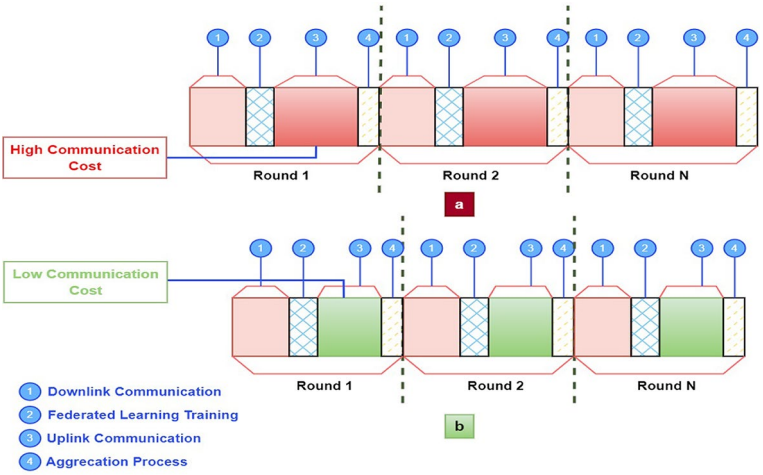


Fig. 8 Communication cost representation

sents the number of rounds and communication differences between high and low communication costs.

The authors of Konečný et al. (2017) proposed two approaches to manage communication cost, such as structured and sketched updates. The structured approach restricted updates such as low rank and random mask in two ways. The sketched update, on the other hand, compressed the trained data before sending to the server. The compression was performed using subsampling, random rotation, and quantization. Similarly, approaches like CMFL, and ADMM (WANG et al. 2019; Zhou and Li 2022) have been used to identify the irrelevant updates and to reduce the client footprint. Such reduction of communication overheads such as irrelevant updates guaranteed learning convergence. The authors of Yao et al. (2019) proposed a features fusion method to manage the federated FedAvg communication cost and performance drops between local and global models. The authors of Dai et al.

(2019) proposed a method for a hyper-sphere quantization framework for achieving a better trade-off between communication and gradient accuracy with different configurations.

The quantization method has been effectively used for compression. Similarly, the adapting FedAvg and Adam optimization has been used to reduce the communication cost using compression techniques. The FedPAQ (Reisizadeh et al. 2020) method was seen to effectively manage the computation using three features such as periodically updating the local device to the server device, involving particle devices for training, quantization of messages to the server. These three parameters have helped address the communication cost and scalability of federated learning. The FetchSGD (Rothchild et al. 2020) method has similarly addressed the communication bottleneck and convergence issues in federated learning using sketching. The momentum and error accumulation were carried out in the sketch. The top-k extracted the information from the local client and broadcasted it. The authors of Hamer et al. (2020) proposed an ensemble-based algorithm for managing communication efficiency with the gradient commutation methods. The ensemble-based algorithm reduced client-to-server and server-to-client communication costs. For reducing the upstream and downstream communication costs in federating learning, the T-FedAvg (Xu et al. 2022a) method was used. The author of Li et al. (2021b) proposed a compressed sensing federated algorithm (CS-FL) to manage downstream and upstream communication between clients and servers. The proposed CS-FL algorithm was compressed with former basic algorithms such as FedAvg and SignSGD.

Using the ternary federated protocol, both the direction data was compressed, and the energy compression reduced. The authors of Nguyen et al. (2021a) proposed a fast convergence algorithm for decreasing the communication rounds and increasing the accuracy of communication. The proposed FOLB algorithm was found to handle communication and heterogeneous devices effectively and to estimate the capabilities of updates. This work increased the model accuracy, convergence speed and model stability across the learning task. Different approaches for the ease of management of the communication burden in IoT systems have been proposed using federated learning with the help of quantization and scarification methods. The authors of Liu et al. (2022a) proposed a method for tackling the communication costs of expensive in-vehicle networks. In another instance, the authors proposed the FedCPF method for customized local training to achieve convergence within a few communication rounds. Similarly, uplink congestion introduced the practice of partial client participation to avoid numerous updates. Compared to the traditional base method (FedAvg) was found to achieve better testing accuracy and communication optimization. In a similar vein, the authors of Wu et al. (2022) proposed an efficient federated learning method for effective communication using knowledge distillation. With all the work done in this field till now in mind, the author's proposed method, called the fedKD framework has been found to effectively reduce communication cost without any notable performance sacrifices. The proposed FedKD exchanges the local updates of a small model, rather than the original model, thus reducing the communication cost using a mentee and mentor relationship. The FedKD framework was evaluated using different scenarios and was found to reduce 94.89% of the communication cost and achieved comparatively better results in a global model. The authors of Asad et al. (2021), Luo et al. (2021a) presented the cost-effective design and evaluating cost-effectiveness for reducing communication costs in federated learning. In the decentralized environment, as the number of edge participants increases, the communication cost also increases significantly due to devices being located in different

geo-locations. To address this issue, 2D-Ring network and 2D-attention-based parameter synchronization and device placement algorithms were introduced (Zong et al. 2024) to reduce the communication overhead in decentralized, federated learning. The summary of communication cost management methods are presented in Table 8

Discussion Due to the huge number of parameters and the continuous updating in federated learning, a huge number of updates are required from the clients to the server and from the server to the clients. This in turn increases communication cost and throughput, thereby shooting up the overall energy requirements. To avoid these issues, effective communication methods are paramount. Various researchers have considered irrelevant updates (WANG et al. 2019), communication bottleneck problems (Rothchild et al. 2020; Liu et al. 2022a), and have reduced the upstream and downstream communication costs (Xu et al. 2022a; Li et al. 2021b). In addition, improvements in convergence speed, effective resource allocation for reducing communication cost and different communication conditions have also been considered to reduce the overall communication cost in federated learning.

8 Applications and functionalities of federated learning

In this section, the various design functionalities of federated learning and how they have been incorporated into different applications in general have been examined closely. Upon some scrutiny, it is evident that sectors like healthcare sectors, industry, insurance, and smart city applications have been early adopters of federated learning. A glimpse of various applications related to federated learning have been provided in the following references (Zhou et al. 2021; Zheng et al. 2022; Nguyen et al. 2022b; Prayitno Shyu et al. 2021).

The federated learning structure prevalent in the intergrated applications is shown in Fig. 9. The figure illustrates how healthcare record data is usually shared using federated learning (Li 2019). The authors of Xu et al. (2021a), Nguyen et al. (2022b) presented a brief literature about how electronic records are shared across the hospital administration. Similarly, the author of Rieke et al. (2020) presented a picture of the future of digital health and highlighted pertinent challenges with respect to using federated learning. In a similar vein, the authors of Qayyum et al. (2021) presented a clustered learning framework for diagnosing COVID-19 using a remote healthcare centre without sharing sensitive data. However, none of these endeavours have chosen to focus on the selection criteria for data and other functionalities of federated learning. Rather, the authors (Yuan et al. 2020a; Choudhury et al. 2020; Li et al. 2022a) chose to present how federated learning is incorporated into the IoT framework and is used to manage the accuracy and communication between the server and decentralized clients. In another instance, the authors of Arikumar et al. (2022), Chen et al. (2021b) proposed a federated learning-based healthcare monitoring system using wearable devices. In this approach, unlabelled data was recovered from wearable devices and was then incorporated into the labelled data and then processed as per subsequent medical processes.

The authors of Hiessl et al. (2022, 2020), Savazzi et al. (2021), Nguyen et al. (2021b), Wang et al. (2022) referred to the present industrial applications that use federated learning. The authors describe the industrial federated system design and the requirements (Hiessl et al. 2020) and exchanges of data that take place without sharing private data. The authors described the different industrial requirements such as machine type, environmental condi-

Table 8 Summary of communication cost reduction methods

S. no.	Methods	Goal	Dataset	Metrics
1	Structured and Sketched updates Konečný et al. (2017)	Reduction in Communication Cost	CIFAR-10	No of Rounds and Accuracy
2	CMFL WANG et al. (2019)	Avoided communication overhead and irrelevant updates	Simulated using Amazon EC2	Communication rounds and Accuracy
3	FEATURE FUSION Method Yao et al. (2019)	Managed Communication cost and performance drop	CIFAR-10	Communication rounds
4	Hyper-sphere Quantization Dai et al. (2019)	Achieved continuum trade-offs	CIFAR-10	Communication rounds and Accuracy
5	Adapting FedAvg and Adam Optimization Mills et al. (2020)	Reduced Communication costs	MNIST/CIFAR-10	Sparsity rates and Total time
6	FedPAQ Reisizadeh et al. (2020)	Addressed Communication and Scalability problems	CIFAR-10, MNIST	Communication time per round
7	FetchSGD Rothchild et al. (2020)	Addressed Communication bottleneck and Convergence issues	CIFAR-100	Overall Compression and Accuracy
8	FedBoost Hamer et al. (2020)	Ensemble learning used for Communication efficiency	Synthetic dataset	Divergence and Communication rounds
9	T-FedAvg Xu et al. (2022a)	Reduced the upstream and downstream Communication cost	MNIST, CIFAR-10	Communication rounds
10	FOLB Nguyen et al. (2021a)	Optimized the Convergence speed	MNIST	Accuracy and Communication cost
11	CS-FL Li et al. (2021b)	Compressed the upstream and downstream data between client server communication in IoT system using FL	MNIST	Accuracy and Communication cost
12	FedCPF Liu et al. (2022a)	Reduced the Communication overhead in edge computing using FL	MNIST, FEMNIST	Accuracy and Communication optimization
13	Communication Efficiency via Knowledge Distillation Wu et al. (2022)	Avoided heavy overheads and high environmental burdens using FedKD	MIND and ADR	Fscore and Accuracy

tions, and operational conditions under which the federated learning cohorts need to operate in terms of establishing and updating the learning tasks. In such cases, the FLaaS has been used to create industrial client devices. Using industrial federated learning, clients work on a demand basis, while the architecture works on an independent structure of ML models and manages non-IID data using cohorts. The FlaaS supports the above conditions. The authors of Wang et al. (2022) proposed industrial personalized federated learning for collaboration between more relevant devices. Federated learning enabled better balance between self-learning and collaborative learning and produced better heterogeneous data for personalization. The authors of Savazzi et al. (2021), Nguyen et al. (2021b) presented the future industrial uses with respect to federated learning. The authors additionally contributed to an understanding of how industrial systems are connected and can cooperate in an automated way to share knowledge and information. The authors of Zheng et al. (2022), Ramu et al. (2022), Jiang et al. (2020) discussed various areas and challenges prevalent in smart cities. In particular, the authors' integrated vehicle system and its corresponding autonomous, communication, and how the incentives were integrated into federated learning were examined. To manage the unlabelled data in smart cities, the authors of Albaseer et al. (2020) proposed a model called Fedsem. Using this work, the labelled data was trained in a global model and injected with unlabelled data into the clients. The authors of Chhikara et al. (2021) proposed the prediction of an Air Quality Index with data privacy to measure the performance of AQI with security features, using UAVs with Long Short-Term Memory (LSTM) and Federated Learning. The authors of Liu et al. (2021a) presented the limitations of the lack of mobility, the crowdsourcing-based monitoring method, and 2-D monitoring using Graph CNN-based long short-term memory (GCLSTM) model with Dense-MobileNet (DM) and Federated Learning techniques to overcome the limitations.

The authors of Long et al. (2020), Shingi (2020), Yang et al. (2019b) presented how federated learning has been used in finance. The authors of Long et al. (2020) presented the management of non-IID data and personalized modelling in the open bank. The authors of Yang et al. (2019b) proposed a federated learning-based credit card fraud detection model using the shared global models across different banks. The shared model was used to make predictions without sharing the cardholder's data. Compared to the traditional models, this model was found to achieve 10% better accuracy.

Discussion The different applications used federated learning concepts for processing data in a decentralized way with privacy protection. In applications, the data was dynamically changed based on the time series. Dynamic data management in federated learning has potential for further research. More specifically, most of the applications' client selection criteria, aggregation conditions, and dynamic updating of knowledge sharing in distributed environments have yet to be addressed by researchers.

9 Challenges and future direction of research in design of federated learning

This section presents the challenges and future research direction in federated learning in terms of functionalities such as node selection, node dropping, aggregation, knowledge transfer, non-IID data management, incentives, and communication cost.

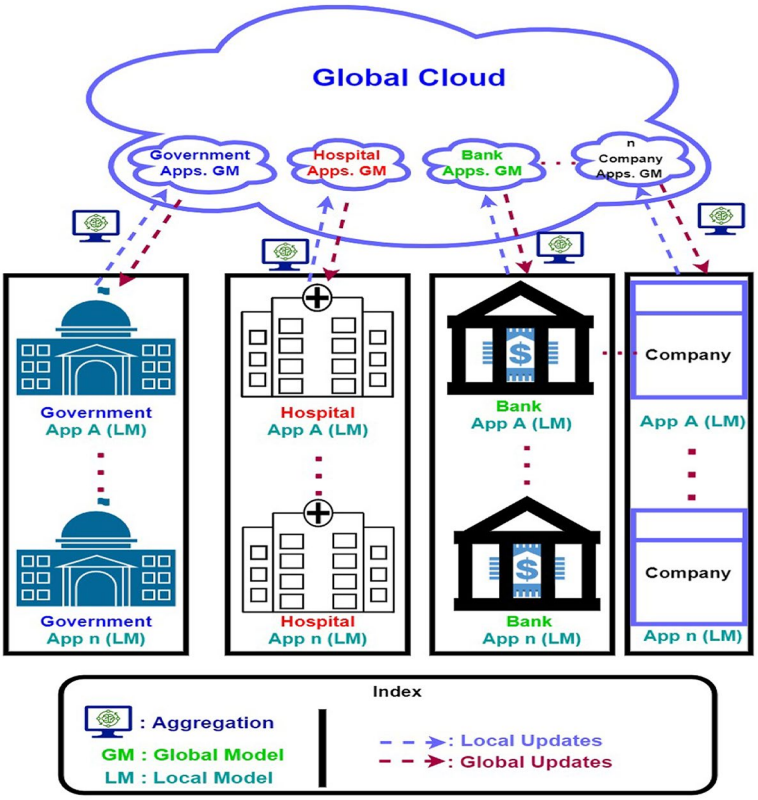


Fig. 9 Integrated applications of federated learning

Node Selection The potential for further research exists in the area of node selection methods in federated learning used to select the client's data in distributed data environments. Aspects that require consideration are client data updating methods, quality of data, client selection criteria (Abdulrahman et al. 2021b), homogeneous data (Nishio and Yonetani 2019), uncertain data management updating protocols (Yoshida et al. 2020), energy management in clients (Zheng et al. 2021), client scheduling (Xu et al. 2021b), removal of noise data (Yang et al. 2022), selection of labelled data on similarly fashioned clients and comparison to nearest clients based on same label (apple to apple comparison).

Aggregation The main goal of the aggregation method is to help increase the learning rate of the global model from distributed clients. Reduction in communication rounds, multi-centre aggregation (Xie et al. 2021), conditions for aggregation (Guha et al. 2019), robust schemes for aggregation (Pillutla et al. 2022; Yeganeh et al. 2020), matching with similar clients prior to aggregation (Wang et al. 2020), aggregation with similar models (Palihawadana et al. 2022), asynchronous aggregation (Nguyen et al. 2022c), Effective aggregation rules on the server side (Ding et al. 2022), and energy consumption during aggregation are some of the pertinent future directions to be considered with respect to aggregation design in federated learning.

Knowledge transfer The main purpose of knowledge transfer is the effective transmission of knowledge for better learning even under varying situations. Promising future directions in knowledge transfer include knowledge distillation (Hinton et al. 2015; He et al. 2020), heterogeneous data features and parameter sharing (Diao et al. 2020), neural architecture for knowledge sharing (He et al. 2021), effective gradient updates (Wang and Charles 2021) and application-wise effective topology development for knowledge sharing. These are some of the most dominant functional points to be considered with respect to knowledge transfer in federated learning.

Data management (Non-IID) In federated learning, data management is used to manage the locally produced data with different or non-uniform data distributions of client participants. Different models have been presented based on different behaviours and requirements of uniform data management. In this context, promising areas of future research include management of privacy during knowledge distillation, management of large client data, data training and probability of data dropping, management of real image data (Luo et al. 2019; Hsu et al. 2020), feature overlapping between the data and clients (Wu et al. 2020), automated machine learning, as well as the influence of managing the distribution of non-IID (Yuan et al. 2020b; Zhu and Jin 2022).

Incentive The main purpose of the incentives in federated learning is to train the model with high-quality data for better accuracy with reduced model iterations. The main challenge of such incentives in federated learning is improved performance. The incentive-based method is required to improve the accuracy and reduce the cost based on the practical incentive-based approaches in federated learning (Zeng et al. 2021).

Communication Cost The primary goal in this case is to reduce the energy used in the entire communication of federated learning. Based on the main challenge the focus should be centred on finding an effective model for efficient federated learning. The key future research directions therefore include reduced-energy green federated learning, use of different models to reduce the communication rounds between clients and the global model, and an effective uplink model, all of which would contribute to reducing the overall communication cost in federated learning.

Scalability Scalability is essential for real-world scenarios. Scalability challenges of federated learning across diverse and extensive networks can be addressed with various strategies such as managing communication overheads, computational load balancing and resource management, Incentive mechanisms and scalability-focused algorithms.

i. Communication overheads can be efficiently reduced by using compression techniques, reducing the waiting time by using asynchronous communication, better bandwidth, designing an effective model with the option to reduce or drop updates, and by increasing the overall model performance (Hafi et al. 2024).
ii. The load can be distributed evenly to other nodes to prevent bottlenecks, and tasks can be transferred from overloaded nodes to unused ones, thereby increasing efficiency (Folino et al. 2024).
iii. Incentives mechanisms for participation and contribution to high-quality updates, thereby improving overall quality, e.g., monetary rewards, reputation systems (Zhou et al. 2023).
iv. Finally, well-designed algorithms that focus on particular tasks to increase scalability, e.g., FedAvg, Fed Prox and Personalised FL algorithms.

10 Conclusion

Federated learning has ushered in a new era for decentralized machine learning. By virtue of its vast applicability, federated learning has garnered a lot of attention from the technical community in the last four years. Since all application data in Fl are privately computed without sharing, it has found eager adoption and endorsement in multiple areas such as healthcare, banking, smart-cites etc. However, in order to assimilate all its benefits, an effective functional design of federated learning is imperative. In this work, all the necessary functional design requirements in FL such as node selection, reason for dropping nodes, effective aggregation, knowledge sharing, data management and communication cost have been duly addressed and presented.

Additionally, all recent research work of significance pertaining to the various functional design aspects of federated learning have been summarized in this work with focus on the design function methods, goals, and any related discussion points. It is the hope of the authors that this research article will help shine the light on new directions of research in the design of federated learning which will, in turn, help to develop newer applications in the area. To that end, the future direction of the research along with the inherent challenges have been summarized as well. Some of the important areas that warrant further research are as follows: i. Effective node selection methods that consider appropriate incentives for data quality for better accuracy. ii. Novel conditions required to inhibit unwanted node selection. iii. Aggregation methods that consider local aggregation to decrease communication cost. iv. Knowledge transfer mechanisms that not only consider transmission from the local to the server, but also the transfer of knowledge or weights to local nodes using sub-global or sub aggregation methods to avoid communication cost. v. A new, green federated learning framework that enables communication between the client to global model rounds with a considerably smaller energy footprint for better and more sustainable computing.

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Declarations

Competing interests The authors declare no competing interests.

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